

Practical: Windows Server Update Services (WSUS)

1. Install and Configure Windows Server Update Services (WSUS)

Step 1: Add WSUS Role

1. Open **Server Manager**.
2. Click **Manage → Add Roles and Features**.
3. Click **Next** on the Before You Begin page.
4. Select:
 - **Role-based or feature-based installation**
5. Click **Next**.
6. Select your server.
7. In the Roles list, check:
 - **Windows Server Update Services**
8. A popup will open → click **Add Features**.
9. Click **Next**.

Step 2: Select WSUS Services

Tick:

- **WSUS Services**
- **Database (WID Database)** (default recommended)
- **WSUS Management Console**

Click **Next**.

Step 3: Select Content Location

Choose folder:

D:\WSUS

or

C:\WSUS

Click **Next**.

Step 4: Install Role

1. Click **Install**.
2. Wait for the installation to complete.
3. Do **NOT** close Server Manager.

Step 5: Post-Installation Configuration

1. Click the **Notification Flag** in Server Manager.
2. Click:

Launch Post-Installation tasks

3. Wait until it finishes.

Step 6: WSUS Configuration Wizard

1. Open:

Tools → Windows Server Update Services

2. In the Configuration Wizard, choose:

Synchronize from Microsoft Update

3. Click **Next**.
4. Set Proxy:
 - If no proxy is required → click **Next**.
5. Choose languages:
 - **English** (or as needed)
6. Choose Products:
 - **Windows 10**
 - **Windows 11**
 - **Windows Server 2019**
7. Choose Classifications:
 - **Security Updates**
 - **Critical Updates**

8. Click **Next**.
9. Click **Start Synchronization**.

2. Create a Group of Windows Client Machines and Configure Windows Update Using Group Policy

Step 1: Open Group Policy Management

1. On the Server, press **Win + R**.
2. Type:

gpmc.msc

3. Press **Enter**.

Step 2: Create New GPO

1. Right-click **Group Policy Objects**.
2. Click **New**.
3. Name:

WSUS Policy

Step 3: Edit Policy

1. Right-click **WSUS Policy**.
2. Click **Edit**.
3. Go to:

Computer Configuration → Administrative Templates → Windows Components → Windows Update

Step 4: Configure WSUS Settings

1. Set WSUS Server

1. Open:

Specify intranet Microsoft update service location

2. Enable it.
3. Enter:

http://<WSUS-SERVER-IP>:8530

Example:

http://192.168.1.10:8530

2. Enable Automatic Updates

1. Open:

Configure Automatic Updates

2. Enable it.
3. Select:

Option 4 - Auto download and schedule install

Step 5: Link GPO to Client OU

1. Open **Active Directory Users and Computers**.
2. Create OU:

ClientMachines

3. Move the client PCs into this OU.
4. Link **WSUS Policy** to this OU.

Step 6: Apply Policy on Client

On the client machine, run:

```
gpupdate /force
```

3. Approve a Windows Security Update for Installation

Step 1: Open WSUS Console

Go to:

Tools → Windows Server Update Services

Step 2: Synchronize Updates

1. Click **Synchronization**.
2. Wait for the updates list.

Step 3: Approve Update

1. Go to:

Updates → All Updates

2. Filter:
 - **Status: Needed**
3. Select one update.

Example:

Security Update for Windows Server

4. Right-click the update.
5. Click **Approve**.

Step 4: Choose Target Group

Select:

All Computers

or create a group such as:

TestGroup

Click:

Approved for Install

Step 5: Client Update Process

On the client:

Run:

wuaclt /detectnow

or

usoclient startscan

Then run:

gpupdate /force

Check:

Settings → Windows Update → Check for updates

4. Generate a WSUS Patch Report

Step 1: Open Reports

In the WSUS Console, go to:

Reports → Update Status Summary

Step 2: Select Report Type

Choose:

Update Status Summary

or

Computer Detailed Status

Step 3: Generate Report

Click:

Run Report

Step 4: Export Report

You have 2 options:

Option A: Export as CSV

1. Click **Export**.
2. Select:
 - **CSV format**
3. Save the file.